Mathematical overview

For a 4-right, 3-up grid displacement using R,U,D, let k be the number of diagonal right-up steps (0<=k<=3). A monotone route then has 4-k right steps and 3-k up steps. If R/U cost 1 and D costs d, total price is 7+(d-2)k. Thus use three diagonals for d<2, none for d>2, and any k for d=2. This is a cost metric; a diagonal's physical Euclidean length is not its printed coin price. With unit-cost diagonals, the minimum number of moves is max(4,3)=4. Diagonal travel changes the base staircase rule.

Monotone R/U routes on a 3-by-3 grid all have six edges, but there are 20 distinct ordered paths. Counts at a vertex equal the sum of counts at its left and lower predecessors; a blocked vertex has count zero. For one blocked vertex (i,j), paths removed equal the number S-to-(i,j) times the number (i,j)-to-F. The printed target counts 8 and 11 arise from blocking (1,1)/(2,2) and (1,2)/(2,1). One nonendpoint vertex cannot block all routes because two outer-boundary paths share only S and F.

With retracing permitted, the fixed narrow strip has no upper bound on finite S-to-F path length. A horizontal out-and-back segment of quarter-unit length at S lies inside |y-x|<=1/4 and adds 1/2 unit each time. A quarter-step monotone staircase inside the strip has length 8. Repeating that loop 44 or 184 times produces lengths 30 and 100. Each path is finite. The unbounded claim uses arbitrarily many finite repetitions, not an infinite physical string or a limit path.

Grades 2-5 explore prices and finite path catalogs. Grades 4-5 explain the price threshold, count with predecessor sums, and distinguish closeness from traveled length under retracing. String comparisons are experiments; step totals, path bijections and repeat constructions prove these claims. Coordinates refer to ideal centerlines, not an allowance from printed stroke thickness.

Entry, preparation, common launch, and flexible use

Unpiloted; rulers, string, counters and print scale have not been physically rehearsed. Page 1 needs following right/up/diagonal moves and counting coins through 10. Page 2 needs a recoverable R/U word and ordered route distinctions. Page 3 needs reading a narrow corridor, adding finite lengths and using a repeat count; the proof can use drawings rather than limits or formal analysis. K-1 can join priced route trials with adult support or return to the base staircase pages; an independent older counting packet is not imposed on them.

Per pair: student pages, ruler, two counters, twelve same-value coins or counters for costs, one removable dot-cover sticker, pencil, and about 50 cm of nonstretch string plus removable tape. Scratch grids or a transparent sleeve allow repeated trials without erasing discoveries. The page-1 grid spacing is 74 points (26.1 mm); page 2 is 80 points (28.2 mm). Their coins count steps, so exact physical step size does not change the price. Page 3 has 30 mm units, a 120 mm square, and a vertical corridor half-width of 7.5 mm. Check the printed 30 mm calibration before geometric measurement. Long repeated routes should be calculated from a short measured loop, not represented by 3 metres of simultaneous string.

For eleven children at K–1 / 2–3 / 4–5: five pair kits; ten route counters, sixty cost coins, five removable blockers, five rulers, five pencils, five 50 cm strings, and tape at three tables. The third older child checks prices or blockers and rotates roles. Keep three adults anchored. A first route uses page 1; counting and backtracking can each occupy a separate return visit.

Launch, 4 minutes: let children move counters along grid edges. Show the one-cell visual: R then U costs 2; D reaches the same corner at its card price. Use a non-task price, such as 4 coins, and let a partner check a short bill. On a counting visit, demonstrate one legal R/U word with a counter and keep it as the one record. On a corridor visit, trace a short legal horizontal piece and its return, counting both traversals. Do not provide the optimal long-loop counts at launch.

Suggested timing: 15-25 minutes for price choices, 20-40 minutes for catalogs/obstacles, 20-35 minutes for corridor designs and explanation. The six-edge catalog can be pooled across pairs. After a short catalog becomes unwieldy, ask children how each final step could have arrived; do not print the recurrence as the method they must follow. Save the chosen blocked vertex with its count.

Solutions, hints, extensions

1. D=1: cheapest price 4, using three D and one R, with four possible orders of those moves. D=2: price 7 for every route, with diagonal counts 0,1,2,3 all possible. D=3: cheapest price 7, using no diagonals; any R/U route works. The formula in the overview proves these optima and that no other diagonal count ties outside d=2. Optional hint: replace one R and one U by one D and compare prices. Extension: choose other widths/heights or asymmetric R/U prices; the diagonal threshold becomes the sum of the two orthogonal prices. Distinguish priced cost from physical length.

2. Full count 20: choose the positions of the three R steps in a six-letter word, or use a 4-by-4 table of predecessor sums. Rows of route counts from y=0 upward are (1,1,1,1), (1,2,3,4), (1,3,6,10), (1,4,10,20). Blocking (1,1) removes 2 x 6=12, leaving 8; blocking (2,2) does the same. Blocking (2,1) or (1,2) removes 3 x 3=9, leaving 11. Complete remaining-count table for nonendpoints: (0,1)=10,(0,2)=16,(0,3)=19,(1,0)=10,(1,1)=8,(1,2)=11,(1,3)=16,(2,0)=16,(2,1)=11,(2,2)=8,(2,3)=10,(3,0)=19,(3,1)=16,(3,2)=10. Two edge routes RRRUUU and UUURRR are vertex-disjoint except at endpoints; one allowed blocker cannot hit both. Optional hint: sort words by their final move, or count routes through the blocked dot. Extension: use two blockers and investigate complete obstruction or a chosen remaining count. All claims here use only right/up movement.

3. Quarter staircase: repeat R(1/4), U(1/4) sixteen times. Its length is 8 and vertical gap from y=x ranges from -1/4 to 0. Before following it, repeat R(1/4), L(1/4) at S 44 times for total 30; repeat it 184 times for total 100. Each repeated segment stays in the same strip and adds 1/2. For any proposed upper bound, choose sufficiently many loops then finish along the staircase. The whole path stays inside the 4-by-4 square, a stronger restriction than the printed strip. Optional hint: keep a legal short piece and ask what happens if it is traversed twice more. Extension: disallow retracing, impose a finite grid resolution, or limit turns; these change the mathematical problem and require a new argument. Never claim an infinite route has been physically built.

Sources, distinction, and verification

Base Week 50 compares equal-length staircases, closeness/turn minima and a finite leftward detour. Page 1 changes the allowed directions and cost metric; page 2 counts/designs paths with obstacles; page 3 allows repeated left/down travel and establishes unbounded finite length inside a fixed corridor.

The organizer/coordinator consulted Math Circle by the Bay, “Paths in Mouseville,” printed pp. 69-73 (PDF pp. 82-86). Its draw/list-first then predecessor-sum format informs optional adult hints; our 3-by-3 blocker board is original, not its map or exact problem. The preface printed pp. viii-x supports return visits. Base source: John D. Norton, Badly Behaved Curves, as identified in the base guide; no new external puzzle copied.

student/verify.py independently enumerates all allowed R/U/D routes for every price, all 20 R/U paths and every single-blocker case, and exact rational corridor vertices, length totals and repeat counts. Checking vertices suffices for full segment containment because y-x is linear on each segment. General optimum and unboundedness proofs appear above. No classroom or physical pretest is claimed.